

Table 62. SPI characteristics⁽¹⁾ (continued)

Symbol	Parameter	Conditions	Min	Typ	Max	Unit
$t_{h(MI)}$	Data input hold time	Master mode	2	-	-	ns
$t_{h(SI)}$		Slave mode	3	-	-	ns
$t_{a(SO)}$	Data output access time	Slave mode	9	-	34	ns
$t_{dis(SO)}$	Data output disable time	Slave mode	9	-	16	ns
$t_{v(SO)}$	Data output valid time	Slave mode $2.7\text{ V} < V_{DD} < 3.6\text{ V}$	-	10	16	ns
		Slave mode $2\text{ V} < V_{DD} < 3.6\text{ V}$	-	10	22	
$t_{v(MO)}$		Master mode	-	3	5.5	ns
$t_{h(SO)}$	Data output hold time	Slave mode $2\text{ V} < V_{DD} < 3.6\text{ V}$	8	-	-	ns
$t_{h(MO)}$		Master mode	1.5	-	-	ns

1. Evaluated by characterization – Not tested in production.
2. Maximum frequency in Slave transmitter mode is determined by the sum of $t_{v(SO)}$ and $t_{su(MI)}$ which has to fit into SCK low or high phase preceding the SCK sampling edge. This value can be achieved when the SPI communicates with a master having $t_{su(MI)} = 0$ while Duty(SCK) = 50%

Figure 24. SPI timing diagram - slave mode and CPHA = 0

